INCH-POUND
MIL-DTL-16377/4B(SH)
26 May 2023
SUPERSEDING
MIL-F-16377/4A(SH)
29 July 1981

DETAIL SPECIFICATION SHEET

FIXTURES, LIGHTING; FLUORESCENT AND SOLID-STATE EQUIVALENT, DETAIL LIGHTING FOR SECRETARY BUREAU, 8 WATTS, 120 VOLTS, 60 HERTZ, SYMBOLS 142.2 AND 142.2SSL

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-16377.

Unless otherwise noted, all dimensions are in inches. For tolerances other than those specified herein, see 3.5.2 of MIL-DTL-16377.

Figures 1 through 5 provide information for the light fixtures covered by this specification sheet. See the requirements sections for specific references to these figures.

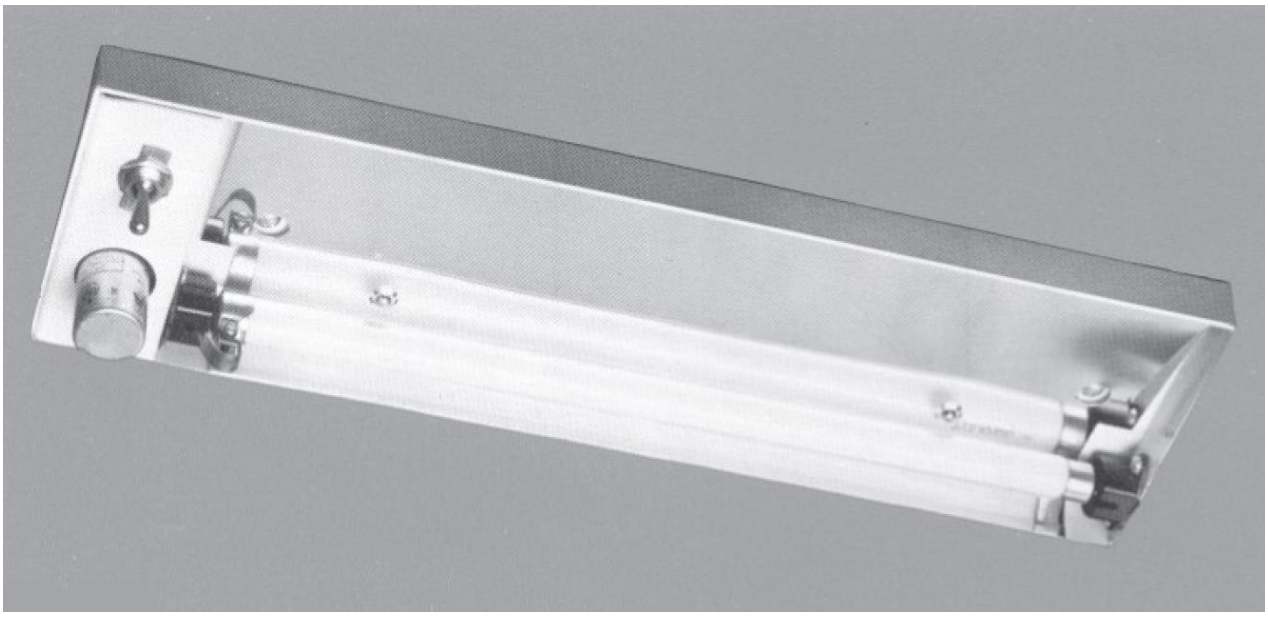


FIGURE 1. Typical view of symbol 142.2.

AMSC N/A FSC 6210

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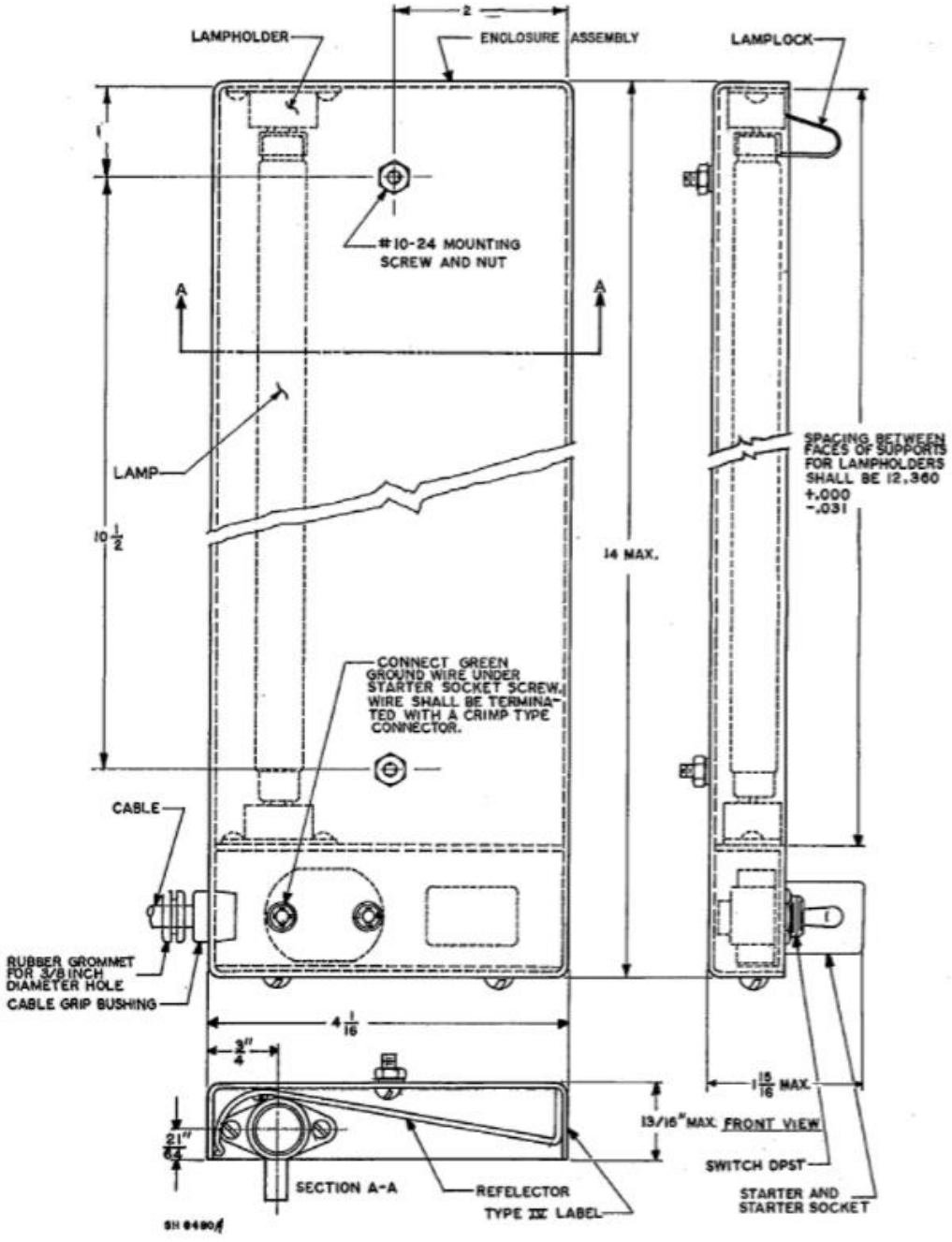


FIGURE 2. Light, fluorescent, detail lighting for secretary bureau, symbol 142.2.

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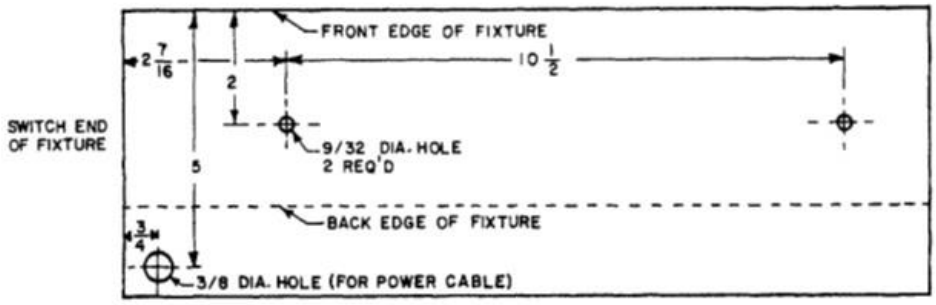


FIGURE 3. Template for mounting.

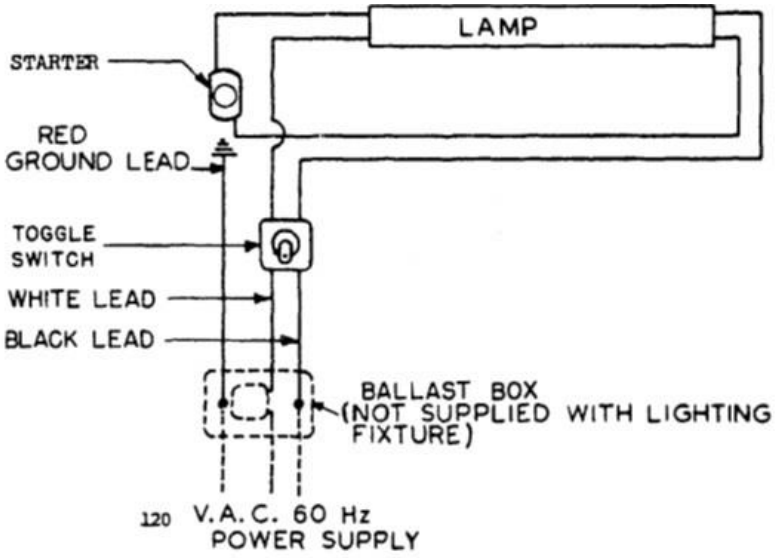
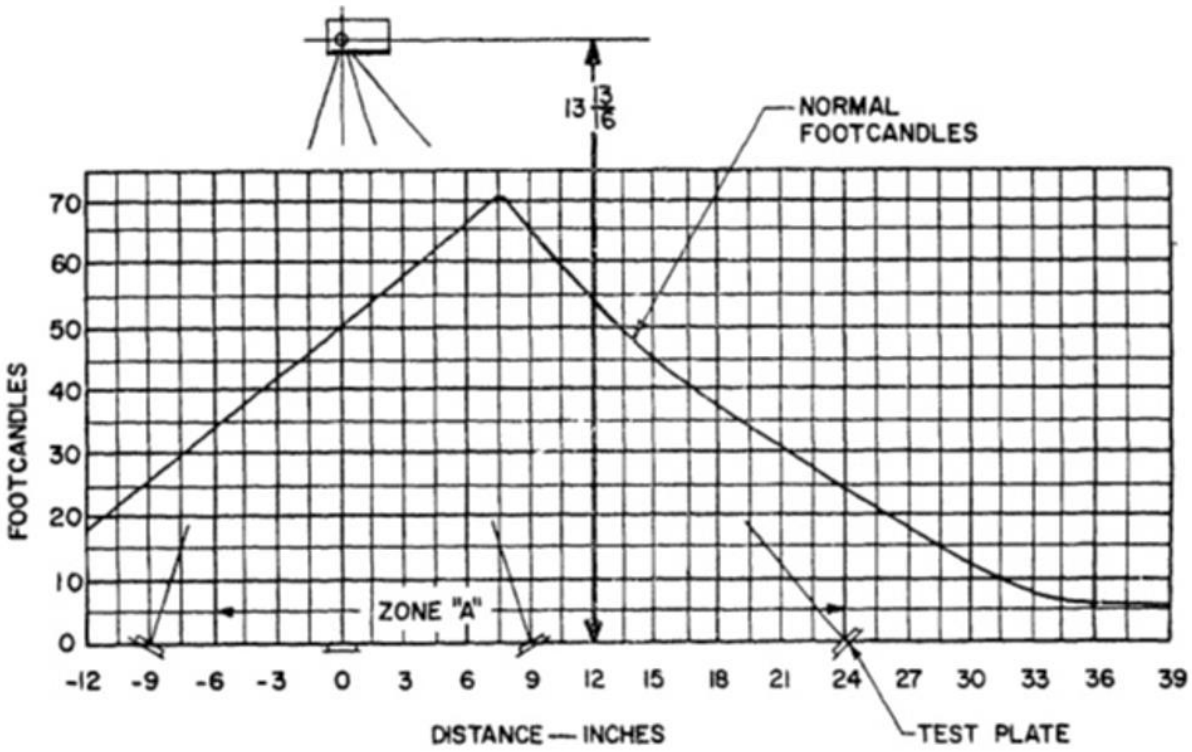


FIGURE 4. Fluorescent wiring diagram.

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Photometric test data:

Distance: $13\frac{13}{16}$ inches.

Voltage: 115 volts, 60 hertz.

Ambient temperature: 76 °F.

Lamp furnished with fixture shall be used.

Ballast, symbol 514 shown on MIL-F-16377/46 shall be used.

FIGURE 5. Footcandle distribution.

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FLUORESCENT REQUIREMENTS

CLASSIFICATION

Type: I
Class: 1

SYMBOL NUMBER.

142.2: Light for secretary bureau, with one 8-watt cool white lamp.

DESIGN

Electrical rating:	8 watts, 120 volts, 60 hertz.
Dimensions and configuration:	See figures 1 and 2.
Template for mounting:	See figure 3.
Cable:	10 feet, TXW-3, in accordance with MIL-DTL-24640/20.
Wiring diagram:	See figure 4.
Reflector:	Specular, aluminum prefinished lighting sheet.
Instruction sheet:	An instruction sheet containing the following information shall be packaged with each fixture.

"THIS FIXTURE IS DESIGNED FOR USE WITH A SEPARATELY MOUNTED 8-WATT BALLAST BOX SHOWN ON MIL-F-16377/46, SYMBOL 514. THE BALLAST BOX IS NOT FURNISHED WITH THE FIXTURE AND MUST BE ORDERED SEPARATELY. FOR INSTALLATION INSTRUCTIONS, SEE NAVSEA DRAWING 803-5184170."

Enclosure:	Aluminum, 0.050 minimum thickness.
Weight:	1 pound, 8 ounces maximum.
Designation and marking:	The following labels shall be furnished in accordance with MIL-DTL-16377 and applied as shown on figure 2. One type I. One type IV. The information to be shown on the type IV label shall be "USE LAMP NSN 6240-00-299-5546."
Standard parts:	The standard parts shown in table I shall be used in the construction of the fixture specified herein.

SOLID-STATE LIGHTING (SSL) REQUIREMENTS

CLASSIFICATION

Type: III
Class: 1

SYMBOL NUMBER.

142.2SSL: Light for secretary bureau, SSL, white.

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DESIGN

Electrical rating:	8 watts, 120 volts, 60 hertz.
Maximum dimensions:	See figure 2.
Wiring:	The wiring shall maintain same control scheme for light output as legacy fluorescent fixture.
Cable:	10 feet, TXW-3, in accordance with MIL-DTL-24640/20.
Photometric data:	See figure 5 of this sheet and 3.7.1 of MIL-DTL-16377. The minimum luminescence shall be not less than that shown on figure 5.
Chromaticity:	See 3.7.4 of MIL-DTL-16377.
Fail-safe circuit:	See 3.6.17 of MIL-DTL-16377.
Weight:	1 pound, 8 ounces maximum.
Designation and marking:	One type I label shall be furnished in accordance with MIL-DTL-16377.
Material:	Corrosion-resistant in accordance with MIL-DTL-16377.

EXAMINATIONS AND TESTS

Examinations and tests shall be as specified in MIL-DTL-16377 and as follows:

First article inspection:	See table II.
Conformance inspection:	
Comparison inspection:	Same as the first article inspection.
Conformance inspection:	See MIL-DTL-16377.

GENERAL INFORMATION FOR FLUORESCENT

Assignment of military part numbers and general information is as follows:

a. Military part number: M16377/4-142.2

b. National Stock Number (NSN): 6230-00-548-0385

c. Supersession data: Formerly covered on NAVSHIPS drawing 9000-S6401-73808 and MS17195.

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TABLE I. Standard parts for type I fixtures.

Name	Specification or Military Part Number	NSN	Remarks
Lamp	W-L-116/3	6240-00-299-5546	Industry number F8T5/CW
Lamplock	M16377/40-001	6250-00-399-7039	---
Lampholder	UL 542	6250-00-377-8590	Supplement SA
Startersocket	UL 542	6250-00-295-2738	Supplement SA
Starter	NEMA ANSI C78.180	6250-00-299-5962	Commercial designation: FS 5
Toggle switch, DPST	MS16569-1	5930-00-719-9659	---

TABLE II. First article and comparison inspection for complete type I fixtures.

Tests	Requirement Paragraph of MIL-DTL-16377	Test Paragraph of MIL-DTL-16377	Remarks
Examination	3.12	4.5	---
Operation	3.12	4.8.1	1/
Dielectric withstanding voltage	3.6.15	4.8.2	---
Insulation resistance	3.6.16	4.8.3	---
Vibration	3.5.13	4.8.8	---
Dielectric withstanding voltage	3.6.15	4.8.2	---
Insulation resistance	3.6.16	4.8.3	---
Noise	3.5.14	4.8.9	1/
Grounding circuit	3.6.13.3	4.8.11	---
Light output	3.7.1	4.8.6.2	Footcandle
Magnetic permeability	3.4.8	4.8.18	---

NOTE:

1/ Lights shall be connected to ballast box shown on MIL-F-16377/46, symbol 514.

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TABLE III. First article and comparison inspection for complete type III fixtures, symbol 142.2SSL.

Tests	Requirement Paragraph of MIL-DTL-16377	Test Paragraph of MIL-DTL-16377	Remarks
Examination	3.12	4.5	---
Operation	3.12	4.8.1	---
Dielectric withstanding voltage	3.6.15	4.8.2	1/
Insulation resistance	3.6.16	4.8.3	1/
Power interface	3.6.18	4.8.21	---
Shock	3.5.12	4.8.7	Grade B 1/
Vibration	3.5.13	4.8.8	1/
Brightness	3.7.5	4.8.6.4	---
Dielectric withstanding voltage	3.6.15	4.8.2	1/
Insulation resistance	3.6.16	4.8.3	1/
Photometrics	3.7.1	4.8.6	---
Chromaticity	3.7.4	4.8.6.5	---
Correlated color temperature, white light only	3.7.4.1.1	4.8.6.6	---
Lumen maintenance	3.7.11	4.8.20	---
Failsafe circuit design	3.6.17	4.8.22	---
Stress relief	3.8.5	4.8.5	---
Magnetic permeability	3.4.8	4.8.18	---
Optical uniformity	3.7.6.1.1.1	4.8.4	---

NOTE:
1/ In accordance with 4.3.1 of MIL-DTL-16377, tests shall be performed on one luminaire in the order specified.

CHANGES FROM PREVIOUS ISSUE. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Preparing activity:
Navy − SH
(Project 6210-2023-004)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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